

Wazuh Agent Setup and Configuration Manual (Ubuntu)

1. Introduction

This document explains:

- Wazuh agent installation on Ubuntu
- Connecting the agent to Wazuh manager
- File Integrity Monitoring (FIM)
- Understanding `ossec.conf`
- Important monitoring options
- Testing and troubleshooting

2. Wazuh Architecture

Components

Component	Purpose
Wazuh Manager	Central server that receives logs/events
Wazuh Agent	Installed on monitored systems
Wazuh Indexer	Stores alerts and events
Wazuh Dashboard	Web UI for monitoring and analysis

3. Ubuntu Agent Installation

Step 1: Download and install agent

```
curl -sO https://packages.wazuh.com/4.x/apt/wazuh-agent_4.8.2-1_amd64.deb  
sudo WAZUH_MANAGER='192.168.1.40' dpkg -i ./wazuh-agent_4.8.2-1_amd64.deb
```

Replace:

```
192.168.1.40
```

with your Wazuh manager IP.

Step 2: Enable and start agent

```
sudo systemctl daemon-reload  
  
sudo systemctl enable wazuh-agent  
  
sudo systemctl start wazuh-agent
```

Step 3: Verify status

```
sudo systemctl status wazuh-agent
```

Expected:

```
active (running)
```

4. Important Wazuh Agent Commands

Start agent

```
sudo systemctl start wazuh-agent
```

Stop agent

```
sudo systemctl stop wazuh-agent
```

Restart agent

```
sudo systemctl restart wazuh-agent
```

Check status

```
sudo systemctl status wazuh-agent
```

View logs

```
sudo tail -f /var/ossec/logs/ossec.log
```

5. Main Configuration File

Configuration file location

```
/var/ossec/etc/ossec.conf
```

Edit using:

```
sudo vim /var/ossec/etc/ossec.conf
```

or

```
sudo nano /var/ossec/etc/ossec.conf
```

6. File Integrity Monitoring (FIM)

What is FIM?

FIM monitors files and directories for:

- File creation
- File deletion

- File modification
 - Permission changes
 - Ownership changes
 - Content changes
-

7. Basic FIM Configuration

Example:

```
<syscheck>
  <disabled>no</disabled>

  <directories realtime="yes">/home/burp/test</directories>
</syscheck>
```

8. Advanced FIM Configuration

Recommended production configuration:

```
<syscheck>
  <disabled>no</disabled>

  <directories check_all="yes"
             realtime="yes"
             report_changes="yes"
             whodata="yes">/home/burp/test</directories>
</syscheck>
```

9. Meaning of Important Options

check_all="yes"

Monitors all file attributes.

Includes:

- MD5 hash

- SHA1 hash
- SHA256 hash
- File permissions
- File owner
- Group
- File size
- Modification time
- Inode

Example

```
chmod 777 file.txt
```

Wazuh detects permission change.

realtime="yes"

Detects changes immediately using Linux inotify.

Without realtime:

- Wazuh waits for scheduled scans.

With realtime:

- Alerts appear instantly.

Example

```
touch file.txt
```

Alert generated immediately.

report_changes="yes"

Stores file content differences.

Useful for:

- Text files
- Configuration files
- Scripts

- Logs

Example

Old content:

```
version1
```

New content:

```
version1  
version2
```

Wazuh can show content differences.

whodata="yes"

Tracks:

- Who modified the file
- Process name
- Process ID
- Username
- Command executed

Requires:

```
sudo apt install auditd -y
```

Enable:

```
sudo systemctl enable auditd  
sudo systemctl start auditd
```

10. What is a Hash?

A hash is a unique digital fingerprint of a file.

Even small file changes create a completely different hash.

Example:

File content:

```
hello
```

SHA256 hash:

```
2cf24dba5fb0a...
```

After modification:

```
hello world
```

New hash:

```
b94d27b9934d3...
```

Wazuh compares hashes to detect changes.

11. Restart Agent After Configuration Changes

Always restart after editing `ossec.conf`.

```
sudo systemctl restart wazuh-agent
```

12. Testing FIM

Create test directory

```
mkdir -p /home/burp/test
```

Create file

```
echo "version1" > /home/burp/test/demo.txt
```

Wait 20 seconds.

Modify file

```
echo "version2" >> /home/burp/test/demo.txt
```

Delete file

```
rm -f /home/burp/test/demo.txt
```

13. Viewing Alerts in Dashboard

Open:

```
https://<WAZUH_SERVER_IP>
```

Go to:

- Threat Hunting
- Security Events

Search:

```
rule.groups:syscheck
```

or

```
syscheck.path:/home/burp/test/demo.txt
```

14. Common FIM Alerts

Event	Meaning
added	File created
modified	File modified
deleted	File deleted
permissions_changed	chmod executed
ownership_changed	chown executed

15. Troubleshooting

Agent not running

Check:

```
sudo systemctl status wazuh-agent
```

Restart:

```
sudo systemctl restart wazuh-agent
```

No alerts appearing

Check:

- Agent connected to manager
- Correct directory path
- Restarted agent
- Dashboard filters

View logs:

```
sudo tail -f /var/ossec/logs/ossec.log
```

whodata not working

Install auditd:

```
sudo apt install auditd -y
```

Enable:

```
sudo systemctl enable auditd  
sudo systemctl restart auditd
```

Restart agent:

```
sudo systemctl restart wazuh-agent
```

report_changes not showing

Requirements:

- Text file
- Small file
- File modified after baseline created
- report_changes enabled

Correct order:

1. Create file
2. Wait 20 seconds
3. Modify file

16. Recommended Production Configuration

```
<syscheck>  
  <disabled>no</disabled>  
  
  <frequency>43200</frequency>  
  
  <scan_on_start>yes</scan_on_start>
```

```
<directories check_all="yes"
           realtime="yes"
           report_changes="yes"
           whodata="yes">/home/burp/test</directories>

<skip_nfs>yes</skip_nfs>
<skip_dev>yes</skip_dev>
<skip_proc>yes</skip_proc>
<skip_sys>yes</skip_sys>

</syscheck>
```

17. Useful Linux Commands for FIM Testing

Create file

```
touch file.txt
```

Modify file

```
echo "hello" >> file.txt
```

Change permissions

```
chmod 777 file.txt
```

Change owner

```
sudo chown root:root file.txt
```

Delete file

```
rm -f file.txt
```

18. Best Practices

Recommended directories

Good choices:

- /etc
- /var/www
- Application configs
- Scripts
- Security-related files

Avoid monitoring in realtime:

- /tmp
 - /proc
 - /sys
 - Large log directories
-

19. Conclusion

Wazuh File Integrity Monitoring helps detect:

- Unauthorized changes
- Malware activity
- Webshell uploads
- Configuration tampering
- Insider threats
- Permission changes
- File deletions

Using:

```
check_all="yes"
realtime="yes"
report_changes="yes"
whodata="yes"
```

provides a powerful real-time security monitoring solution for Linux systems.